

An AI-Ready Air-to-Ground Channel Model for HAPS-UAV Access Networks

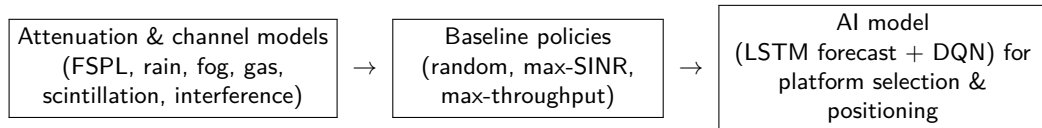
Harsha Matadhikari

Online PG Diploma in Advanced Communication Engineering with Quantum AI Integration
Indian Institute of Technology Delhi

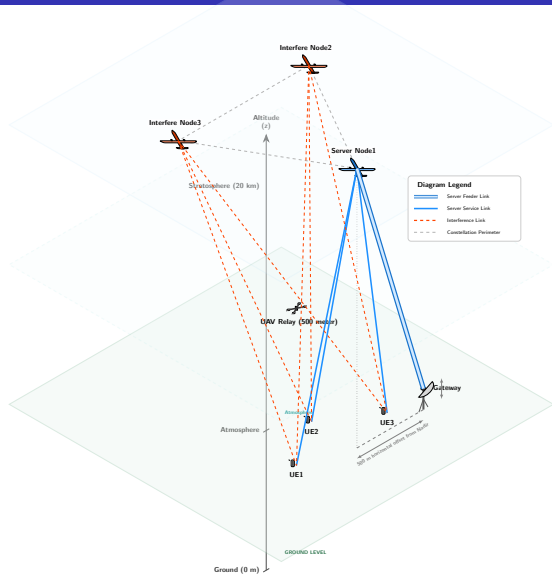
Motivation

- A 3-HAPS equilateral-triangle constellation serves a ground footprint via **service** (2 GHz), **backhaul** (38 GHz), and two-hop **UAV relay** links.
- Distance-only link budgeting misses a key effect: co-channel interference from the other two HAPS makes SINR depend strongly on *angular position*, not just distance.
- Static link-quality analysis cannot react to this, or to time-varying rain fades on the 38 GHz backhaul.
- **AI Ready Model:** a validated channel model, an LSTM rain forecaster, and a DQN agent that learns real-time user–platform association.

Research flow:



System Topology and Channel Model



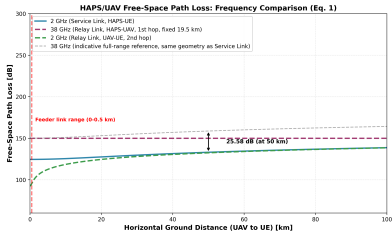
Three link types:

- HAPS $\rightarrow$ UE (service, 2 GHz)
- HAPS $\rightarrow$ Gateway (backhaul, 38 GHz)
- HAPS $\rightarrow$ UAV $\rightarrow$ UE (relay; feeder hop at 38 GHz, UAV $\rightarrow$ UE access hop at 2 GHz)

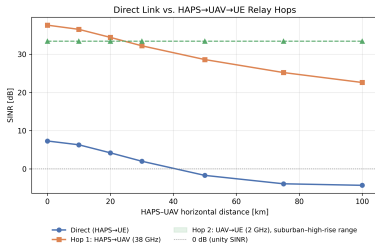
Channel model:

- Free-space path loss (HAPS tier)
- Building-blockage LoS/NLoS probability (UAV tier, Holis & Pechac 2008)
- ITU-R rain, fog, gas, and scintillation losses (backhaul/feeder)

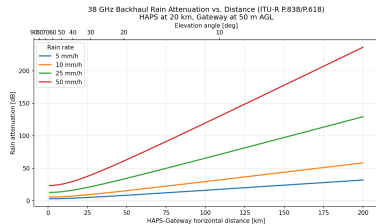
Path Loss and SINR vs. Distance



FSPL vs. distance – same form for 2 GHz and 38 GHz; the feeder sits ≈ 25.6 dB higher purely from the frequency term.



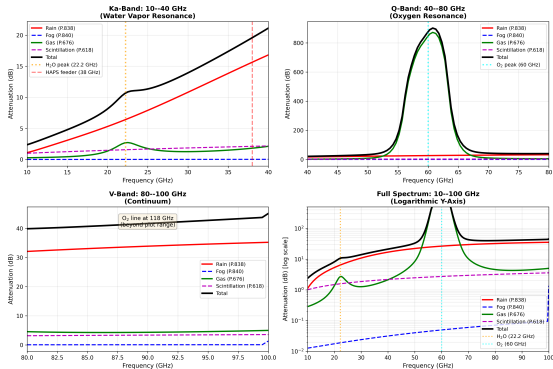
SINR vs. distance – direct link falls below 0 dB past ≈ 40 km, while both relay hops (feeder 38 GHz, access 2 GHz) stay well above it at every distance tested.



Rain attenuation vs. distance (38 GHz backhaul) – grows faster than linear past ≈ 100 km as elevation drops and the rain slant-path lengthens.

Why 38 GHz: Frequency Selection

Atmospheric Loss vs Frequency — Nominal (50 km horizontal)
10 mm/h Rain, 21.75° Elevation, Delhi



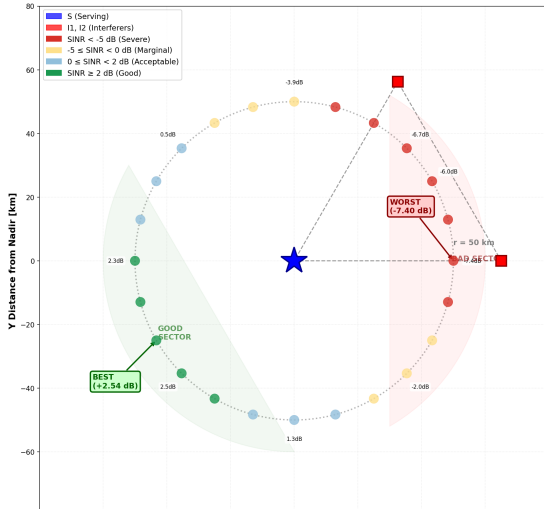
Atmospheric loss (rain, fog, gas) swept across the Ka/Q/V bands for the Delhi gateway path, 10 mm/h rain:

- **Rain** (P.838/P.618) – dominant term, grows roughly linearly with frequency; up to ≈ 19 dB at 99.9% availability ($R_{0.1} = 15.7$ mm/h).
- **Fog/cloud** (P.840) – small except in dense fog; negligible below 10 GHz.
- **Gas absorption** (P.676) – two resonance peaks to avoid: H₂O at 22.2 GHz, O₂ at 60 GHz (there, total loss exceeds 800 dB over 50 km – unusable).

38 GHz sits in the valley between these peaks (≈ 0.2 – 0.3 dB/km clear-sky gas loss) – this, not just available bandwidth, is why it was chosen over a frequency closer to either resonance.

Angular Interference Sectors and Scintillation Fading

HAPS Constellation Geometry with Spatial SINR Sectors
User positions at $r = 50$ km | Circle color = SINR value
Red=Bad (-7.40 dB) | Yellow=Marginal | Green=Good (+2.54 dB)



Angle-dependent SINR (sectors): the 3-HAPS triangle creates a good sector ($\approx 120^\circ$ – 290° , away from interferers) and a bad sector ($\approx 330^\circ$ – 75° , toward them) – same radius, up to 9.94 dB apart.

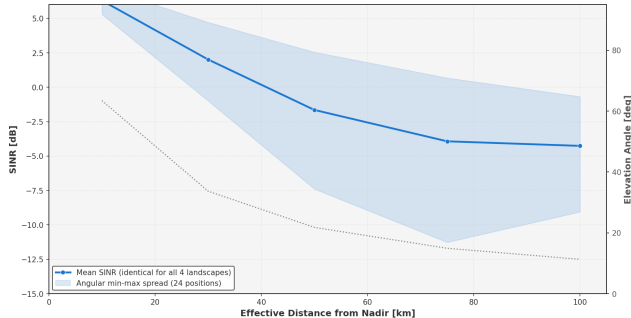
Scintillation (ITU-R P.618): rapid amplitude fluctuations from tropospheric turbulence; grows with frequency, shrinks with elevation. At 30° elevation, $p = 0.01\%$: ≈ 0.49 dB at 38 GHz vs. ≈ 0.09 dB at 2.1 GHz.

Favors positioning the UAV relay near nadir (higher elevation) when feasible, especially for the HAPS–UAV feeder hop – which also tends to be away from the interferer-facing sector.

Nuance: each HAPS could simultaneously be a *serving* node (near-nadir, low-loss for its own users) and an *interferer* for the other two HAPS' users – so this isn't a single clean optimization across the constellation (future work, see Conclusion).

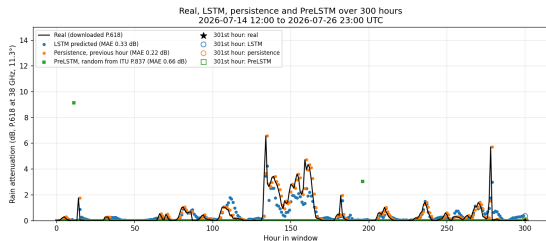
Why Learning is Needed: Spatial SINR Heterogeneity

Figure 5: Multi-HAPS Interference - Signal Power Attenuation Across 100 km Service Footprint
Mean SINR (landscape-independent, S:III.C); shaded band = spatial variation (min-max) across 24 angles around the fixed 65 km equilateral 3-HAPS constellation (interferer geometry, not landscape)



- At a **fixed 50 km radius**, SINR varies by **up to 9.94 dB** depending on angular position relative to the two interfering HAPS.
- For comparison, simply *doubling* the link distance only costs about 2–3 dB – so angular position affects SINR more than distance does.
- A simple “always pick max-SINR” rule cannot anticipate these angular patterns or adapt as weather and battery state change.
- **Conclusion:** a learned, real-time association policy is needed.

LSTM Rain-Attenuation Forecaster



- LSTM predicts feeder (38 GHz) rain attenuation up to **six hours ahead** from NASA POWER / METAR / ERA5 weather history for Delhi.
- More accurate than persistence at 4–6 h (RMSE); all six forecast horizons feed the DQN's state.
- **Observation:** simply averaging the LSTM and persistence forecasts beats both alone on RMSE (0.511 vs. 0.645 / 0.530) – ensembling is a promising future direction.
- DQN must decide at hour t *before* that hour's rain is observed $\Rightarrow$ needs a forecast.
- Evaluated **out-of-sample** (e.g. 2026 forecasts from a model trained on 2023–2025 data).
- In deployment this would be a **continuous, incremental** feed – the model keeps updating as new weather data arrives.

DQN Association Agent: Problem Formulation

Per-user MDP – each user independently chooses a platform; network reward is computed after all assignments.

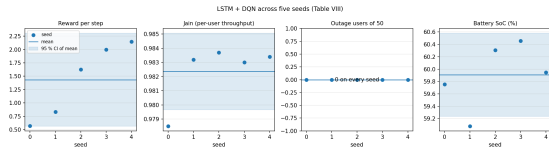
- **State:** SINR to each platform, distance, battery SoC, rainfall, visibility, time-of-day, six LSTM rain forecasts.
- **Action:** $a_k \in \{0, \dots, 5\}$ – direct link to HAPS1–3, or relay through a UAV served by HAPS1–3.
- **Reward:** throughput (rain-degraded) + fairness (Jain index) – energy cost – outage/degraded-service penalties – low-battery penalty.
- Custom Deep Q-Network implemented directly in PyTorch (`torch/torch.nn`).
- Trained on 400 rolling 24-hour episodes from 2024; evaluated on 50 held-out 2025 days, five experiments (random seeds).

Results: Policy Comparison

Policy	Reward/step	Jain	Outage /50
Random (six actions)	−40.5 (1.7)	0.563 (0.007)	16.5 (0.4)
Max-SINR (direct only)	−14.5 (2.1)	0.866 (0.002)	0.9 (0.2)
Max-throughput (six actions)	−3.8 (1.7)	0.988 (0.000)	0.0 (0.0)
LSTM + DQN	+ 1.4 (1.0)	0.981 (0.006)	0.0 (0.0)

- Mean $\pm$ std. dev. over five experiments, 50 held-out 2025 days.
- DQN has the **highest reward**, zero outage, and fairness statistically indistinguishable from max-throughput.
- Max-SINR assumes perfect SINR knowledge and cannot use the relay actions – the DQN doesn't need either assumption.
- Max-throughput also assumes perfect, instantaneous knowledge of every option's SINR/capacity, and has no memory, battery, or fairness awareness – a theoretical upper bound, not a deployable policy.

Results: Stability Across Experiments



LSTM + DQN, five independent seeds (dot = one

training run; line = mean; band = 95% CI of the mean):

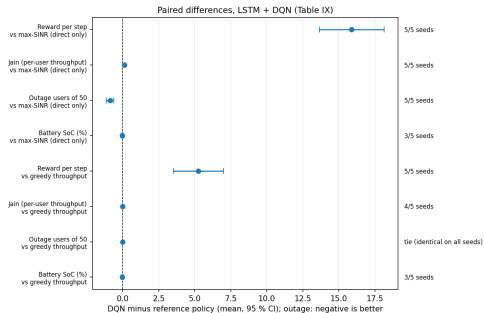
- Reward/step: $+1.40 \pm 0.96$ (95% CI [0.21, 2.59])
- Jain fairness: 0.981 ± 0.006
- Outage: 0.0 ± 0.0 (every seed, every day)
- Battery SoC: $59.96 \pm 0.59 \%$

CI vs. spread: the shaded band is *not* “most runs land here” – it narrows with more seeds even if run-to-run variability (± 0.96) stays the same. It shows how precisely the *true mean* is known from only 5 runs.

On the seed axis: seed is just a run label (0–4 are RNG seeds, not a meaningful scale) – any apparent rise with seed number is run-to-run noise, not a real trend; e.g. Jain and battery SoC don’t follow the same order as reward.

Takeaway: zero outage and tight fairness/SoC variance across all 5 seeds show the policy is **reliably learned**, not a lucky single run.

Results: Statistical Significance



Paired differences, LSTM+DQN minus baseline (per experiment, 50 days each). Whiskers crossing 0 = not significant.

vs. Max-SINR (direct links only):

- Reward: +15.87 ($p < 0.001$) – **significant**, 5/5
- Jain: +0.115 ($p < 0.0001$) – **significant**, 5/5
- Outage: -0.87 ($p = 0.0007$) – **significant**, 5/5
- Battery SoC: +0.03 ($p = 0.44$) – not significant (tie)

vs. Max-throughput (greedy, all 6 actions):

- Reward: +5.17 ($p = 0.001$) – **significant**, 5/5
- Jain: -0.007 ($p = 0.07$) – not significant (tie)
- Outage: 0 – tie, identical on every seed
- Battery SoC: ≈ 0 – not significant (tie)

Takeaway: DQN beats Max-SINR on 3 of 4 metrics. Against the stronger, information-privileged Max-throughput baseline, it wins only on reward – and *ties, never loses*, on fairness, outage, and battery.

Conclusion and Future Work

Channel model. Spatial SINR heterogeneity (up to 9.94 dB at fixed 50 km) dominates service-link quality; backhaul remains noise-limited. **Relay path.** HAPS→UAV→UE outperforms the direct link at every distance tested (11.1 vs. 2.7 bits/s/Hz at nadir); keeping the UAV near nadir (high elevation) on the 38 GHz feeder hop limits rain and scintillation losses, both of which grow sharply as the slant path lengthens at low elevation. **Learned association.** The LSTM-fed DQN beats max-throughput and max-SINR on reward, with zero outage and no significant fairness loss.

Future work:

- Learn *when* relaying is worthwhile as a function of user distance.
- Model movement of the HAPS and UAV platforms.
- Consider more UAVs, including UAV selection.
- Extend the LSTM forecast to neighbouring regions (e.g. ConvLSTM on gridded reanalysis data, or a spatio-temporal GNN if future sites are irregularly placed).
- Adapt to changing HAPS roles (shift a node between service and interferer based on its distance to the gateway).
- Jointly optimize UAV/HAPS positioning across both roles a node could play at once – serving node for its own users, interferer for the other HAPS' users – rather than each in isolation.
- Consider beamforming between HAPS and Gateway.
- Explore forecast ensembling (e.g. LSTM + persistence averaging already improved RMSE in our initial check, slide 5).

Thank you – Questions?